UNIVERSITY EXAMINATION BOARD

ADVANCED CALCULUS & DIFFERENTIAL EQUATIONS

Examination Year: 2026 | Maximum Marks: **100** | Total Questions: **20** | Time Allowed: **3 Hours**

GENERAL INSTRUCTIONS:

1. All questions are compulsory unless specified otherwise.
2. Candidates must show all working, intermediate steps, and derivations to earn full marks.
3. Draw diagrams wherever necessary; label all parts clearly.
4. Figures, graphs and diagrams are provided with the respective question.
5. Use of scientific calculator is permitted unless otherwise notified.

Q. No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Total
Marks	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	100

Question 1 [5 Marks]

Find the general solution to the first-order linear differential equation: $(dy / dx) + 2xy = x e^{-x^2}$.

Question 2 **[5 Marks]**

Compute the directional derivative of f(x,y,z) = x^2 y + y^2 z at point (1,2,3) in the direction of **v** = (1,1,1).

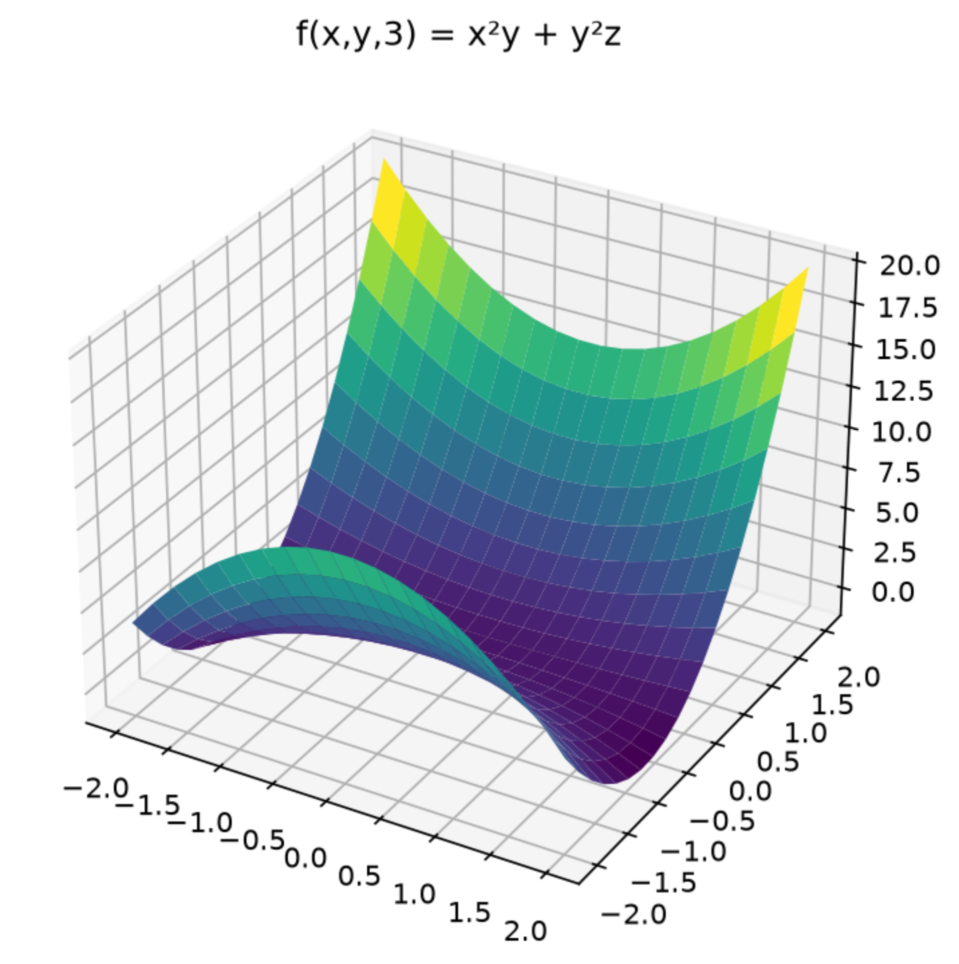

Figure 1

Question 3 **[5 Marks]**

Solve the exact differential equation: (2xy + y^3)dx + (x^2 + 3xy^2)dy = 0.

Question 4 **[5 Marks]**

Find the critical points of f(x,y) = x^3 + y^3 - 3xy and classify them as maxima, minima or saddle points.

Question 5 [5 Marks]

Compute the line integral $\int_C (x^2 + y^2)ds$ where C is the right half of the unit circle $x^2 + y^2 = 1$ traversed counterclockwise.

Question 6 [5 Marks]

Solve the Bernoulli equation: $(dy / dx) + (y / x) = x^2 y^3$.

Question 7 [5 Marks]

Find the Fourier series representation of $f(x) = x$ on the interval $-\pi < x < \pi$.

Question 8 [5 Marks]

Compute the surface integral $_S (x^2 + y^2)dS$ where S is the surface $z = \sqrt{(x^2 + y^2)}$ for $0 \leq z \leq 1$.

Question 9 [5 Marks]

Solve the initial value problem: $y'' + 4y' + 4y = e^{-2x}$, $y(0) = 1$, $y'(0) = 0$.

Question 10 [5 Marks]

Find the general solution to the system of differential equations: $(dx / dt) = 2x + y$, $(dy / dt) = -x + 4y$.

Question 11 [5 Marks]

Compute the divergence and curl of the vector field $\mathbf{F} = (x^2 y, y^2 z, z^2 x)$.

Question 12

[5 Marks]

Solve the partial differential equation: $(\partial u / \partial t) = k (\partial^2 u / \partial x^2)$ with boundary conditions $u(0,t) = u(L,t) = 0$ and initial condition $u(x,0) = f(x)$.

Question 13

[5 Marks]

Find the Laplace transform of $f(t) = t^2 \sin(3t)$.

Question 14

[5 Marks]

Compute the volume integral _V $(x^2 + y^2 + z^2)\, dV$ where V is the unit sphere $x^2 + y^2 + z^2 \leq 1$.

Question 15

[5 Marks]

Solve the Cauchy-Euler equation: $x^2 y'' - 3xy' + 4y = 0$.

Question 16

[5 Marks]

Find the Green's function for the differential operator $L[y] = y'' + y$ with boundary conditions $y(0) = y(\pi) = 0$.

Question 17 **[5 Marks]**

Compute the flux of **F** = (x, y, z) through the surface $z = 1 - x^2 - y^2$ for $z \geq 0$.

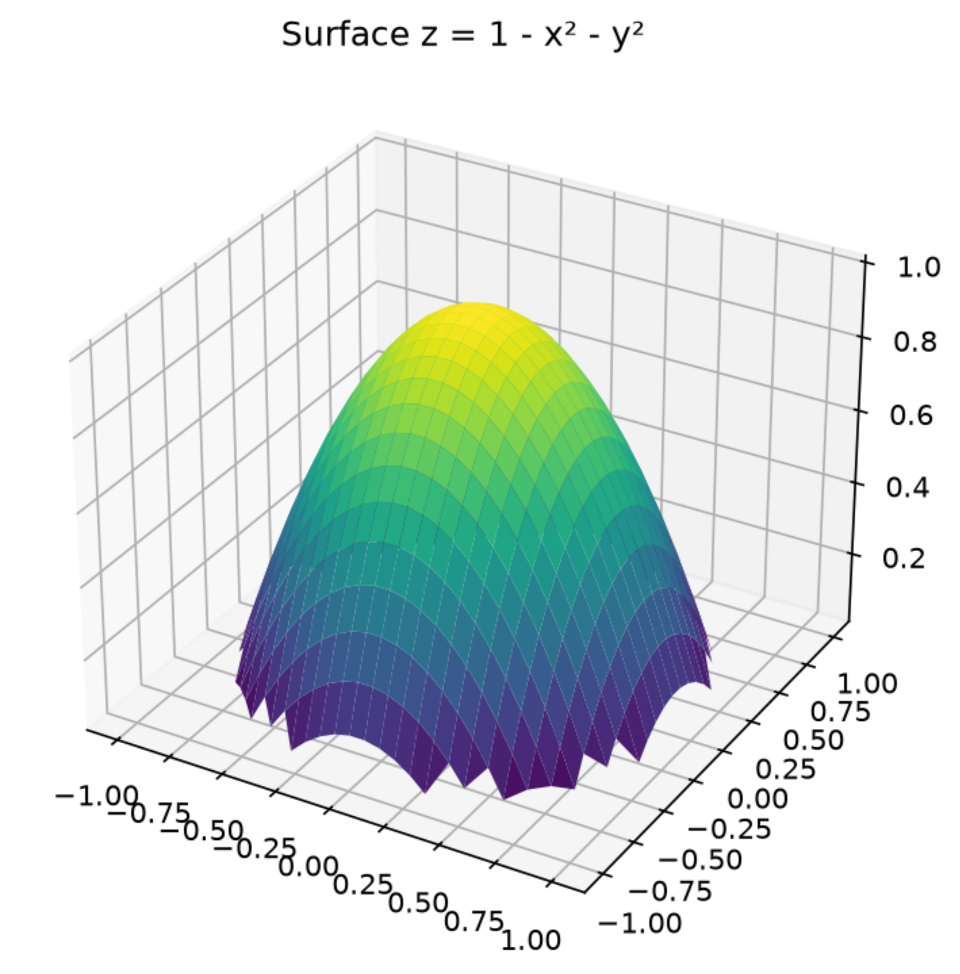

Figure 2

Question 18 **[5 Marks]**

Solve the wave equation $(\partial^2 u / \partial t^2) = c^2 (\partial^2 u / \partial x^2)$ with initial conditions $u(x,0) = \sin(\pi x/L)$ and $(\partial u / \partial t)(x,0) = 0$.

Question 19 **[5 Marks]**

Find the general solution to the nonhomogeneous differential equation: $y'' + y = \tan x$ using variation of parameters.

Question 20

[5 Marks]

Compute the complex line integral ■_C $(e^z / z^2 + 1)\,dz$ where C is the circle $|z| = 2$ traversed counterclockwise.